



basic education

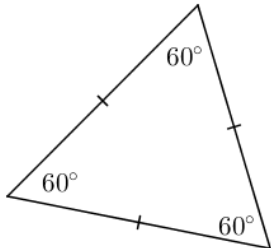
Department:
Basic Education
REPUBLIC OF SOUTH AFRICA

GEC PILOT STUDY MARKING GUIDELINE 2023 MATHEMATICS PAPER 2 GRADE 9

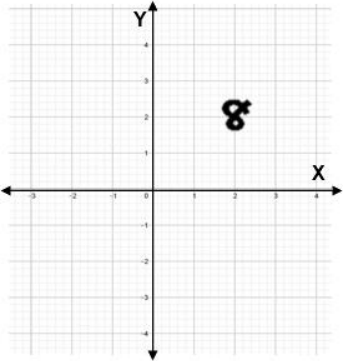
SECTION A

- One mark per answer.
- There are no half marks.

No.		Expected answer		Clarification
1.	C	It is vertically opposite to 88° .	✓	
2.	D	90°	✓	Lines are perpendicular.
3.	C	Corresponding angles.	✓	
4.	B	55°	✓	
5.	A	54°	✓	$3x + 2x = 90^\circ \dots \text{compl } \angle s$ $5x = 90^\circ$ $\frac{5x}{5} = \frac{90^\circ}{5}$ $x = 18^\circ$ $\widehat{EOB} = 3x = 3(18^\circ)$ $\widehat{EOB} = 54^\circ$
6.	B	66°	✓	$68^\circ + \widehat{E} + 46^\circ = 180^\circ \dots \text{sum of } \angle s \text{ of a } \Delta$ $\widehat{E} = 180^\circ - 68^\circ - 46^\circ$ $\widehat{E} = 66^\circ$ $\widehat{E} = f \dots \text{alt } \angle s; EB \parallel DC$ $\therefore f = 66^\circ$ OR $\widehat{B} + \widehat{C} = 180^\circ \dots \text{co-int } \angle s; EB \parallel DC$ $68^\circ + 46^\circ + f = 180^\circ$ $f = 180^\circ - 114^\circ$ $\therefore f = 66^\circ$
7.	A	10°	✓	$20^\circ + 55^\circ + 2a - 15^\circ = 180^\circ \dots \text{co-int } \angle s;$ $MO \parallel PQ$ $2a + 60^\circ = 180^\circ$ $2a = 120^\circ$ $a = 60^\circ$

No.		Expected answer	Clarification
			$2b + 2a + 20^\circ + a - 40^\circ = 180^\circ$...sum of $\angle$ s of a Δ $2b + 2(60^\circ) + 20^\circ + 60^\circ - 40^\circ = 180^\circ$ $2b + 120^\circ + 20^\circ + 60^\circ - 40^\circ = 180^\circ$ $2b = 180^\circ - 160^\circ$ $2b = 20^\circ$ $\frac{2b}{2} = \frac{20^\circ}{2}$ $b = 10^\circ$
8.	A	one right angle and two acute angles.	✓
9.	A		✓
10.	A	The diagonals are perpendicular, one diagonal bisects the other and one diagonal bisects the angles at the vertices.	✓
11.	D	A square is a rectangle.	✓
12.	B	95°	$\hat{A} + \hat{B} + \hat{C} = 180^\circ$...sum of $\angle$ s of a $\Delta = 180^\circ$ $\hat{A} + 47^\circ + 38^\circ = 180^\circ$ $\hat{A} = 180^\circ - 47^\circ - 38^\circ$ $= 95^\circ$
13.	A	162°	$\hat{N}_1 = \hat{P}_1 = 35^\circ$...alt $\angle$ s ; $MN \parallel PQ$ $\hat{N}\hat{O}\hat{Q} = \hat{P}_1 + \hat{P}\hat{N}\hat{O}$...ext. $\angle =$ sum of 2 opp int $\angle$ s $= 35^\circ + 127^\circ$ $\therefore y = 162^\circ$ OR $y = \hat{M}\hat{N}\hat{O}$...alt $\angle$ s ; $MN \parallel PQ$ $y = 35^\circ + 127^\circ$ $y = 162^\circ$
14.	A	4 cm	$\Delta ABC \parallel \Delta EFG$...corr. $\angle$ s = $\frac{AB}{EF} = \frac{BC}{FG} = \frac{AC}{EG}$...corr. sides are proportional $\frac{y}{8} = \frac{6}{12}$

No.		Expected answer		Clarification
				$y = \frac{6 \times 8}{12}$ $\therefore AB = 4 \text{ cm}$
15.	A	115°	✓	$\widehat{ECD} = \widehat{DEC} = y + 15^\circ \dots \angle s \text{ opp} = \text{sides}$ $y + 15^\circ + y + 65^\circ = 180^\circ \dots \angle s \text{ on a straight line}$ $2y + 80^\circ = 180^\circ$ $2y = 100^\circ$ $y = 50^\circ$ $\widehat{E_1} = y + 65^\circ$ $= 50^\circ + 65^\circ$ $= 115^\circ$ OR $\widehat{B} = \widehat{D} = y \dots \text{opp. } \angle s \text{ of a rhombus} =$ $\widehat{E_1} = \widehat{C_1} + \widehat{D} \dots \text{ext. } \angle = \text{sum of 2 opp int } \angle s$ $y + 65^\circ = y + 15^\circ + y$ $y = 50^\circ$ $\widehat{E_1} = y + 65^\circ$ $= 50^\circ + 65^\circ$ $= 115^\circ$ OR $\widehat{B} = \widehat{D} = y \dots \text{opp. } \angle s \text{ of a rhombus} =$ $\widehat{ECD} = \widehat{DEC} = y + 15^\circ \dots \angle s \text{ opp} = \text{sides}$ $\widehat{ECD} + \widehat{DEC} + \widehat{D} = 180^\circ \dots \text{sum of } \angle s \text{ of } \Delta$ $y + 15^\circ + y + 15^\circ + y = 180^\circ$ $3y = 150^\circ$ $y = 50^\circ$ $\widehat{E_1} = y + 65^\circ$ $= 50^\circ + 65^\circ$ $= 115^\circ$
16.	A	80°	✓	$\Delta RMS \equiv \Delta TMS \dots \text{given}$ $\therefore \widehat{RMS} = \widehat{TMS} \dots \Delta s \text{ congruent}$ $x - 20^\circ = 60^\circ$ $x = 80^\circ$
17.	D	16 cm	✓	In ΔNRQ and ΔNOM: $\widehat{Q_1} = \widehat{M_2} \dots \text{corr. } \angle s; QR \parallel MO$ $\widehat{R_1} = \widehat{O_1} \dots \text{corr. } \angle s; QR \parallel MO$ $\widehat{N} = \widehat{N} \dots \text{common. } \angle \text{ or sum } \angle s \text{ of } \Delta$ $\Delta NRQ \parallel \Delta NOM \quad \angle \angle \angle$ $\frac{NR}{NO} = \frac{RQ}{MO} = \frac{NQ}{NM} \dots \text{corr. sides are proportional}$ $\frac{9}{18} = \frac{8}{MO} = \frac{6}{12} \therefore MO = 16 \text{ cm}$ OR

No.	Expected answer	Clarification
		$\hat{M}_1 = \hat{O}_1 \dots \text{alt.} \angle s ; NO \parallel MP$ $\hat{R}_1 = \hat{O}_1 \dots \text{corr.} \angle s ; MO \parallel QR$ $\therefore \hat{M}_1 = \hat{R}_1$ $\hat{P} = \hat{N} \dots \text{opp.} \angle s \text{ of a parm}$ $\hat{M}_2 = \hat{O}_2 \dots \text{alt.} \angle s ; NO \parallel MP$ $\hat{M}_2 = \hat{Q}_1 \dots \text{corr.} \angle s ; MO \parallel QR$ $\therefore \hat{O}_2 = \hat{Q}_1$ $\triangle NRQ \parallel \triangle PMO \quad \angle \angle \angle$ $PO = QN + QM = 12 \text{ cm} \dots \text{opp. sides of a parm}$ $NR = RO = 9 \text{ cm} \dots \text{given}$ $\text{In } \triangle PMO \text{ and } \triangle NRQ$ $\frac{MO}{RQ} = \frac{PM}{NR} \dots \text{corr. sides are proportional}$ $\frac{MO}{8} = \frac{18}{9}$ $MO = 2 \times 8$ $MO = 16 \text{ cm}$
18.	C 	✓
19.	D Q and R	✓
20.	C Translated 2 units to the left and 4 units up.	✓
21.	A $P(-2 ; 7)$	✓
22.	C $Q'(1 ; 5)$	✓
23.	B $(x; y) \rightarrow (-x ; y - 1)$	✓
24.	D $A'(9 ; -5)$ and $B'(2 ; 1)$	✓
25.	D $\triangle KLM$	✓
26.	B 15 cm	✓ $5 \times 3 = 15 \text{ cm}$
27.	B $A = lb$	✓

No.		Expected answer		Clarification
28.	B	the perimeter of a circle.	✓	
29.	D	$A = \pi r^2$	✓	
30.	B	15,88 cm	✓	$A = \pi r^2$ $198 = \pi r^2$ $r^2 = \frac{198}{\pi}$ $r = \sqrt{\frac{198}{\pi}}$ $\text{Diameter} = \sqrt{\frac{198}{\pi}} \times 2 = 15,88$
31.	D	90	✓	$AB + BC + (FE + DC) + (AF + ED)$ $= 20 + 25 + 20 + 25$ $= 90$
32.	C	48 cm ²	✓	$TS = (30 \div 10) = 3 \text{ cm}$ $QS = 6 \text{ cm} \dots \text{PR bisects QS}$ $\text{Area of PQRS} = \frac{1}{2}(\text{PR} \times \text{QS})$ $= \frac{1}{2}(16 \times 6) = 48 \text{ cm}^2$ $= 48 \text{ cm}^2$
33.	B	530 cm ²	✓	$\text{Radius} = 30 \div 2 = 15 \text{ cm}$ $A = \frac{3}{4}\pi r^2$ $A = \frac{3}{4}\pi(15)^2$ $A = 530 \text{ cm}^2$
34.	B	20x cm	✓	$0,08x \text{ m} = 8x \text{ cm}$ $\frac{8x}{2} = 4x \text{ and } \frac{6x}{2} = 3x \dots \text{diagonals bisect at right angles}$ $\therefore EF^2 = (3x)^2 + (4x)^2 \dots \text{Pyth theorem}$ $EF^2 = 9x^2 + 16x^2$ $EF^2 = 25x^2$ $EF = \sqrt{25x^2}$ $EF = 5x$ $\text{Perimeter} = (5x \times 4)$ $\text{Perimeter} = 20x \text{ cm}$
35.	D	13 times	✓	$r = 5 \div 2 = 2,5 \text{ cm}$ $\text{Perimeter of wheel} = 2\pi r$ $= 2\pi(2,5)$ $= 15,71 \text{ cm}$ $2 \text{ m} = 200 \text{ cm}$ $\text{Number of times the wheel turns}$ $= 200 \div 15,71$ $= 12,73074475 \approx 13 \text{ times}$

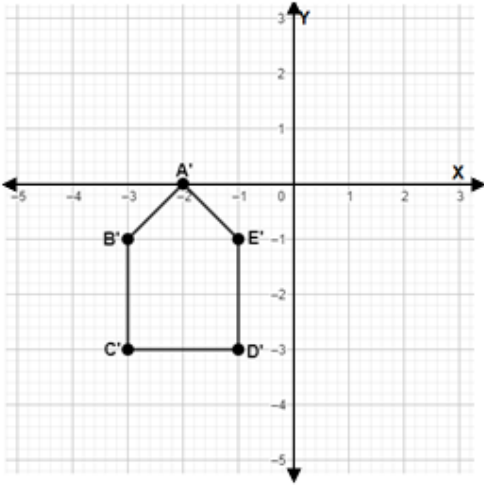
No.		Expected answer		Clarification
36.	C	40 mm	✓	$V = s^3$ $64 = s^3$ $s = 4 \text{ cm} = 40 \text{ mm}$
37.	D	248 cm ²	✓	$SA = 2[(6 \times 4) + (10 \times 4) + (10 \times 6)]$ $= 248 \text{ cm}^2$
38.	A	1120 cm ²	✓	$SA = 2\left(\frac{1}{2} \times 8 \times 15\right) + (25 \times 15) + (17 \times 25)$ $+ (25 \times 8)$ $= 120 + 375 + 425 + 200$ $= 1120 \text{ cm}^2$
39.	B	$SA = 2\pi r^2 + 2\pi rH$	✓	
40.	C	250 cm ³	✓	$0,05 \text{ m} = 5 \text{ cm}$ $V = 5 \times 5 \times 10$ $= 250 \text{ cm}^3$
41.	B	216 l	✓	$400 \text{ mm} = 40 \text{ cm}$ and $600 \text{ mm} = 60 \text{ cm}$ $V = 90 \times 40 \times 60$ $= 216\,000 \text{ cm}^3$ $= 216 \text{ l}$
42.	B	1,242 m ³	✓	$90 \text{ cm} = 0,9 \text{ m}$ H of triangular prism $= 1,4 - 0,9 = 0,5$ $V = (0,9 \times 0,9 \times 1,2)$ $+ \left(\frac{1}{2} \times 0,9 \times 0,5 \times 1,2\right)$ $= 0,972 + 0,27$ $= 1,242 \text{ m}^3$
43.	A	3,28 m ²	✓	$\text{Radius} = 25 \text{ cm} = 0,25 \text{ m}$ $\text{Diameter} = 25 \times 2 = 50 \text{ cm} = 0,5 \text{ m}$ $SA = (0,5)(2,4) + [\pi(0,25)^2 + \pi(0,25)(2,4)]$ $= 1,2 + [0,20 + 1,88]$ $= 3,28 \text{ m}^2$
44.	B	8,98%	✓	$V = \pi(90)^2(197)$ $V = 5\,013\,039,4 \text{ cm}^3$ $5\,013\,039,4 \text{ cm}^3 = 5\,013 \text{ l}$ $\text{Percentage of water in tank:}$ $\frac{450}{5\,013} \times 100 = 8,98\%$
45.	B	150 mm	✓	h^2 of triangle $= (\sqrt{4100})^2 - (40)^2$ $= 4100 - 1600$ $= 2500$ $h = 50 \text{ mm}$ $\text{Area of base} = \frac{1}{2} \times 80 \times 50$

No.		Expected answer		Clarification
				$= 2000 \text{ mm}^2 = 20 \text{ cm}^2$ $300 \text{ ml} = 300 \text{ cm}^3$ $20 \text{ cm}^2 \times H = 300 \text{ cm}^3$ $H = 15 \text{ cm} = 150 \text{ mm}$ OR $h^2 \text{ of triangle} = (\sqrt{4100})^2 - (40)^2$ $= 4100 - 1600$ $= 2500$ $h = 50 \text{ mm} = 5 \text{ cm}$ Area of base $= \frac{1}{2} \times 8 \times 5$ $= 20 \text{ cm}^2$ $300 \text{ ml} = 300 \text{ cm}^3 \text{ (1 ml = 1 cm}^3\text{)}$ $20 \text{ cm}^2 \times H = 300 \text{ cm}^3$ $H = 15 \text{ cm} = 150 \text{ mm}$
				Section A total
				45

SECTION B

- Do not penalise the learner for the same mistake twice.
- There are no half marks.

MARKING CODES	
M	Method mark
A	Accuracy mark
CA	Consistent Accuracy mark

No.	Expected answer	Clarification	Mark
46.	 ✓✓✓M	Translating 5 units to the left: 1 mark Translating 1 unit up: 1 mark Correct quadrant: 1 mark	3
47.	$\hat{O}_1 = x$...vert opp $\angle$s ✓M $\hat{O}_3 = 3x$ $\hat{O}_5 = x$ $3x + 3x + x + x + x + x = 360^\circ$...angles around point ✓M $10x = 360^\circ$ ✓M $\frac{10x}{10} = \frac{360^\circ}{10}$ $x = 36^\circ$ ✓A OR $\hat{O}_3 = 3x$...vert opp $\angle$s ✓M $3x + x + x = 180^\circ$...$\angle$s on a str line AD ✓M $5x = 180^\circ$ ✓M $\frac{5x}{5} = \frac{180^\circ}{5}$ $x = 36^\circ$ ✓A	Statement and reason: 1 mark Statement and reason: 1 mark Simplification: 1 mark Answer: 1 mark OR Statement and reason: 1 mark Statement and reason: 1 mark Simplification: 1 mark Answer: 1 mark Accept any other straight line used.	4
48.	$40 \text{ mm} = 4 \text{ cm}$ ✓A $A = \frac{1}{2}bh$ $100 = \frac{1}{2} \times 4 \times h$ ✓M $100 = 2h$ $h = 50 \text{ cm}$ ✓A	Correct conversion: 1 mark Substitution into the correct formula: 1 mark Answer: 1 mark	3

No.	Expected answer	Clarification	Mark
49.	$\hat{P} + 38^\circ + 75^\circ = 180^\circ \dots \text{sum of } \angle\text{s of } \Delta$ ✓M $\hat{P} = 180^\circ - 113^\circ$ $\hat{P} = 67^\circ$ ✓CA $\hat{T}_1 + 67^\circ = 180^\circ \dots \text{co-int } \angle\text{s PQ} \parallel \text{ST}$ ✓M $\hat{T}_1 = 180^\circ - 67^\circ$ $\hat{T}_1 = 113^\circ$ ✓✓CA OR $\hat{R}_1 = 75^\circ \dots \text{vert opp } \angle\text{s}$ ✓M $\hat{S} = \hat{Q} \dots \text{alt } \angle\text{s; PQ} \parallel \text{ST}$ ✓M $\hat{S} = 38^\circ$ ✓A $\hat{T}_1 = \hat{R}_1 + \hat{S} \dots \text{ext } \angle = \text{sum of 2 opp int } \angle\text{s}$ ✓M $\hat{T}_1 = 75^\circ + 38^\circ$ $\hat{T}_1 = 113^\circ$ ✓CA OR $\hat{P} + \hat{Q} + \hat{R}_2 = 180^\circ \dots \text{sum of } \angle\text{s of } \Delta$ ✓M $\hat{P} + 38^\circ + 75^\circ = 180^\circ$ $\hat{P} = 180^\circ - 113^\circ$ $\hat{P} = 67^\circ$ ✓CA $\hat{T}_2 = \hat{P} \dots \text{alt } \angle\text{s; PQ} \parallel \text{ST}$ ✓M $\hat{T}_2 = 67^\circ$ $\hat{T}_1 + \hat{T}_2 = 180^\circ \dots \angle\text{s on a str line}$ ✓M $\hat{T}_1 + 67^\circ = 180^\circ$ $\hat{T}_1 = 180^\circ - 67^\circ$ $\hat{T}_1 = 113^\circ$ ✓CA	Statement and reason: 1 mark Answer: 1 mark Statement and reason: 1 mark Answer: 2 marks Statement and reason: 1 mark Statement and reason: 1 mark Answer: 1 mark Statement and reason: 1 mark Answer: 1 mark OR Statement and reason: 1 mark Answer: 1 mark Statement and reason: 1 mark Statement and reason: 1 mark Answer: 1 mark Accept any other correct responses.	5

No.	Expected answer	Clarification	Mark
50.	OU = TO = 5 cm ...diagonals of a parm bisect each other ✓M/A In ΔTON $ON^2 = TO^2 - TN^2$...Pyth theorem ✓M $= (5)^2 - (4)^2$ ✓M $= 25 - 16$ ✓M $= 9 \text{ cm}^2$ ON = 3 cm ✓CA	Statement and reason: 1 mark Statement and reason: 1 mark Substitution: 1 mark Simplification: 1 mark Answer: 1 mark	5
51.	Volume of the room $V = l \times b \times h$ $V = 5 \times 3 \times 2$ $V = 30 \text{ m}^3$ ✓A Volume of the box $V = l \times b \times h$ $V = 10 \times 6 \times 4$ $V = 240 \text{ cm}^3$ ✓A $V = 0,00024 \text{ m}^3$ ✓A Number of boxes = $\frac{30}{0,00024}$ Number of boxes = 125 000 ✓CA OR Volume of the room $V = l \times b \times h$ $V = 5 \times 3 \times 2$ $V = 30 \text{ m}^3$ ✓A $V = 3000000 \text{ cm}^3$ ✓A Volume of the box $V = l \times b \times h$ $V = 10 \times 6 \times 4$ $V = 240 \text{ cm}^3$ ✓A Number of boxes = $\frac{3000000}{240}$ Number of boxes = 125 000 ✓CA	Answer: 1 mark Answer: 1 mark Conversion: 1 mark Answer: 1 mark OR Answer: 1 mark Conversion: 1 mark Answer: 1 mark Answer: 1 mark Accept any other correct responses.	4

No.	Expected answer	Clarification	Mark
52.	Angle formed by the ray of light to the top of the flag pole and the horizontal line from the light source to the pole, is common. ✓M Angle between the wooden beam and the horizontal line from the light source to the pole = angle formed by the flag pole and the horizontal line from the light source = 90°. ✓M ∴ Angle formed by the ray of light and wooden beam = angle formed by the ray of light and the flag pole. ✓M The triangles formed by the light source, wooden beam and flag pole are similar. $\frac{10}{110} = \frac{18}{x} \text{ ✓M}$ $10x = 1980$ $\frac{10x}{10} = \frac{1980}{10}$ $x = 198 \text{ from the height of the light source ✓M}$ $h = 198 + 2$ $h = 200 \text{ m ✓A}$	Identifying the common angle: 1 mark Identifying the angles = 90°: 1 mark Concluding 3rd angles =: 1 mark Proportion: 1 mark Answer: 1 mark Answer: 1 mark	6
Section B total			30